

Karmugilan Panchamoorthy

PhD Candidate – Structural Engineering

Department of Civil Engineering

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Research Profile

Research scholar in Structural Engineering specialising in the behaviour of cold-formed steel (CFS) shear wall systems under in-plane shear loading. Research spans large-scale experimental testing, finite element modelling, and experimental–numerical integration using full-scale specimens and 3D Digital Image Correlation (DIC). Demonstrated capability in developing validated numerical models in Abaqus and conducting cyclic loading experiments under established protocols. Strong research interest in sustainable construction and seismically resilient steel structural systems.

Areas of Research Interest

- Behaviour and design of steel structures and structural connections
- Cold-formed steel (CFS) shear wall systems under seismic and cyclic loading
- Finite element method (FEM) based nonlinear simulation using Abaqus
- Experimental–numerical integration using full-scale structural testing
- 3D Digital Image Correlation (DIC) for full-field deformation measurement
- Sustainable and seismically resilient structural systems

Education

Doctor of Philosophy (PhD) - Structural Engineering 2021 - Present

Indian Institute of Technology Hyderabad (IIT Hyderabad), India

Thesis: *[Add thesis title here]*

Supervisor: Prof. Mahendrakumar Madhavan (Professor, Dept. of Civil Engineering)

CGPA: **9.75 / 10**

Master of Engineering (M.E.) - Structural Engineering 2019 - 2021

ACGCET, Karaikudi, Tamil Nadu (Anna University), India

CGPA: **9.71 / 10** *[Add rank / distinction if applicable]*

Bachelor of Engineering (B.E.) - Civil Engineering 2015 - 2019

Government College of Engineering, Thanjavur (Anna University), India

CGPA: **8.44 / 10**

Research Experience

Research Scholar (PhD) Aug 2021 - Present

Dept. of Civil Engineering, IIT Hyderabad — Supervisor: Prof. Mahendrakumar Madhavan

- Designed and conducted large-scale quasi-static cyclic experiments on plywood-sheathed CFS shear wall systems; tested [N] full-scale specimens covering key parametric variables.
- Performed full-field strain and deformation measurement using 3D Digital Image Correlation (DIC), enabling detailed failure mechanism identification beyond conventional instrumentation.

- Developed and validated nonlinear finite element models in Abaqus, achieving close agreement with experimental load–displacement responses and failure modes.
- Integrated experimental and numerical datasets for comprehensive evaluation of shear strength, ductility, and energy dissipation of CFS shear wall systems.
- Investigated screw fastener behaviour in CFS stud-to-sheathing connections, contributing to improved design guidelines for connection performance under shear loading.
- Authored and co-authored peer-reviewed publications in leading structural engineering journals and international conference proceedings (see Publications).

Publications

Journal Articles

- [1] **Panchamoorthy, K. S.** and Madhavan, M. M. “In-Plane Seismic Performance of Double-Sided Steel-Sheathed Cold-Formed Steel Shear Walls: Experimental Evaluation and Design Implications”. In: *ASCE: Journal of Structural Engineering* (2026). DOI: 10.1061/JSENDH/STENG-15872.
- [2] **Panchamoorthy, K. S.** and Madhavan, M. M. “Behaviour of plywood-sheathed cold-formed steel shear walls under monotonic and cyclic in-plane shear loading”. In: *Journal of Constructional Steel Research* 238 (2026), p. 110172. DOI: 10.1016/j.jcsr.2025.110172.
- [3] **Panchamoorthy, K. S.** and Madhavan, M. M. “Experimental Study of Screw Connections in Plywood-Sheathed CFS Wall Panels Subjected to In-Plane Shear Loading”. In: *ASCE: Journal of Structural Engineering* 151.5 (2025). DOI: 10.1061/JSENDH.STENG-14106.

Conference Proceedings

- [1] **Panchamoorthy, K.** and Madhavan, M. “Behaviour of Screw Fasteners in Cold-Formed Steel Stud-to-Sheathing Connections”. In: *Proceedings of the SDSS 2025 International Colloquium on Stability and Ductility of Steel Structures*. Vol. 8. 6. 2025, pp. 953–959. DOI: 10.1002/cepa.70186.

Core Skills

Experimental

- Large-scale and full-scale structural testing under cyclic and quasi-static loading protocols
- 3D Digital Image Correlation (DIC) setup, calibration, and post-processing
- Instrumentation: LVDTs, load cells, strain gauges, data acquisition systems

Numerical and Computational

- Nonlinear finite element modelling and analysis using Abaqus (material and geometric nonlinearity)
- Model calibration, mesh sensitivity studies, and validation against experimental data
- Parametric studies and post-processing of simulation results

Programming and Software

- Programming: Python (data processing, plotting, automation)
- Structural analysis: STAAD.Pro
- Design codes: AISC, Eurocode, IS Codes (CFS-specific: AISI S400, AS/NZS 4600)
- Office / documentation: L^AT_EX, MS Office

Awards & Fellowships

- [Fellowship name] – Ministry of Education / MHRD, Govt. of India 2021 – Present
- [Add any institute medals, best paper awards, scholarships]

Teaching & Mentoring Experience

- **Teaching Assistant** – [*Course Name*], IIT Hyderabad [Semester, Year]
Responsibilities: tutorials, grading, laboratory sessions.
- **Project Mentoring** – Co-supervised [N] B.Tech / M.Tech students on [topic]. [Year]

Invited Talks & Workshops Attended

- [*Talk title / Workshop name*], [Organisation / Conference], [Location] [Year]
- SDSS 2025 – International Colloquium on Stability and Ductility of Steel Structures 2025

Professional Affiliations

- Student Member, [Professional Society, e.g., ASCE / IStructE / ISNT] [Year – Present]

References

Prof. Mahendrakumar Madhavan
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